

Embedded software engineer - 12 years of experience - Swiss work permit c

About

Embedded software engineer with experience in automotive, security and medical domains. Strong expertise in embedded software design, low-level development, debugging and code quality. Proficient in C/C++, Python and communication protocols (TCP/IP, UDP, CAN). Experienced with microcontroller systems, functional safety (ISO 26262), performance constraints and traceability requirements.

Core skills

- Embedded development in C, C++, Python
- C++/Qt application development (desktop)
- CAN bus communication, debugging and system integration
- Software architecture, memory management and optimization
- Code analysis, testing and validation
- Continuous integration, Git workflows, code reviews
- Root cause analysis, hardware debugging and support
- Project coordination (Scrum, Agile)

Technologies

- Methods: Agile, SAFE, Scrum
- Modeling: Scilab, Matlab/Simulink, Grafcet
- Protocols: TCP, UDP, IP, Ethernet, CAN
- Tools: Git, GitHub Actions, Jenkins CI, Subversion, WinDbg, GDB, CodeWarrior, Qt
- PCB design: KiCad
- English C1 – TOEIC 920/990
- French, native

Embedded software expertise

- Languages: C, C++, C#, Python, Java, Shell, VHDL, Verilog
- Testing: unit/integration tests (TESSY, Unity), functional tests (NI TestStand), CI pipelines
- Coding standards: MISRA C, PEP8, GitFlow workflow, pull-request reviews
- GUI frameworks: LVGL (embedded UI), Qt (desktop tools)
- FreeRTOS: tasks, queues, semaphores, ISRs, timers

Microcontrollers & peripherals

- MCU: STM32F4, ESP32, PIC 16F/18F, AVR, 8051, NXP S12Z
- Peripherals: UART, I2C, SPI, CAN, DMA, PWM, ADC, watchdog, timers, flash memory
- Debug tools: JTAG, GDB, WinDBG, oscilloscopes, logic analyzers

Experience

- Oct. 2025 – **Software engineering consultant**, *embarque.ch customer*, Geneva (GE).
Present
- Structuring development tasks and milestones
 - Creation of user and developer documentation (Sphinx)
 - Code reviews and enforcement of code quality and consistency
 - Python code cleaning, PEP8 conformity and dead-code removal
 - Definition of smoke tests and automated test suite
 - Full redesign of the GitHub structure, templates and project documentation
- Jun. 2023 – **Embedded Software Engineer**, *Swatch Group*, Moutier (BE).
Feb. 2024
- Software support for two industrial control products
 - Memory leak analysis and fixes on ESP32 (ESP-IDF, FreeRTOS, OPC-UA, JTAG)
 - NTP synchronization improvements
 - Writing and executing functional test cases
 - Software packaging to standardize on-site deployment
 - UI feature integration using LVGL on LPC1788
- Nov. 2018 – **Embedded Software Engineer**, *LEM*, Meyrin, Switzerland.
Jul. 2022
- Development of current-sensor application (ISO 26262, AUTOSAR, ASPICE, codebeamer ALM)
 - Bootloader development with self-update and external SPI flash management
 - Code reviews and MISRA C compliance
 - Unit and functional testing (TESSY, TestStand) integrated into CI (Jenkins)
 - CAN bus debugging (CANoe, CANstress)
 - Scrum Master for the software team
- May 2016 – **Test Engineer**, *Stormshield*, Lyon, France.
Oct. 2018
- Development of automated test platform for endpoint security product (SES)
 - Daily multi-OS test execution on virtual environments
 - Security testing in Python/C (ACL, registry, firewall, HIPS)
 - Scrum Master for the test team
- Jul. 2013 – **Software Engineer**, *Vossloh Cogifer Monitoring & Telecom*, Lyon, France.
Apr. 2016
- Developed a shared XML parser for both desktop and embedded systems.
 - Designed and implemented a scalable configuration tool (Qt/C++, custom grammar, CSV parameter database).
 - Created a multi-threaded maintenance and monitoring tool (SNMP, parallel system updates/control).
 - Implemented Active Directory data synchronization.
- Jan. 2012 – **Embedded Software Engineer**, *Centum Adeneo*, Lyon, France.
Jun. 2013
- Refactoring for generic reuse across multiple boards (Cypress PSoC)
 - 1-Wire driver for calibration data retrieval
 - IRIG driver with low-pass filter for signal demodulation
- Feb. – Jun. 2011 **Embedded Software Engineer**, *M.G.I. Digital Graphic Technology*, Paris, France.
- Analysis of IP solutions for Xilinx Spartan-6
 - High-speed Ethernet/IP/UDP stack development
 - FPGA updater over IP (C# tool, integrity checks, fail-safe procedures)
- Jan. – Jun. 2010 **Embedded Software Engineer**, *Thalès Trixell*, Moirans, France.
- Integrated Wi-Fi on Xilinx Virtex-5 FPGA SoC (evaluation of IPs and HW solutions).
 - Implemented USB 2.0 and PCIe interfaces on FPGA platform.
 - Set up embedded Linux (DENX) and performed USB performance benchmarks (netperf).

Education

- 2018 **Training**, *IRA – CIPEN*, Paris, Safety and security of industrial control systems.
- 2005–2011 **Engineering Degree**, *Grenoble INP – Esisar*, Valence, Specialization in networks and embedded systems.